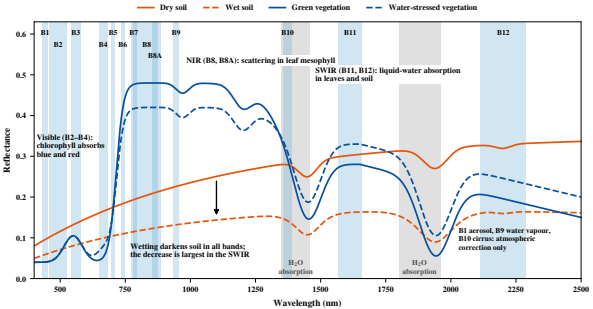
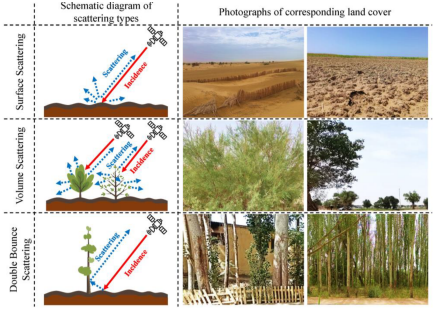
Motivation Different ways of obtaining soil moisture

Source	How SM is obtained	Depth	Main limitation
Microwave (SMAP, SMOS, ASCAT, S1)	invert: σ^0 or $T_B = f(\text{SM}, \text{roughness}, \text{vegetation})$	top $\sim$5 cm; informs part of the root zone (Feldman et al., 2023)	shallow; assumes vegetation and roughness; coarse (9–40 km) except S1
Optical / thermal (S2, Landsat, MODIS)	empirical relations with surface properties: reflectance and indices; LST–vegetation space (e.g. triangle method, TVDI)	skin / surface	indirect, site-specific calibration; clouds; canopy hides the soil
Land-surface models	$d\theta/dt = P - ET - Q - D$, with parameters p	full profile	forcing errors, soil and vegetation parameters, process representation (Koster et al., 2009; Seneviratne et al., 2010)
Reanalysis (ERA5-Land)	LSM driven by reanalysis meteorology	full profile	model assumptions dominate deep layers with few observational constraints (Li et al., 2021; Miguez-Macho and Fan, 2021)
In situ (ISMN)	direct probe measurement	several depths	point scale; sparse, uneven global coverage
Data-driven (ML / DL)	SM $= f_\theta(\text{inputs})$, learned from in-situ labels; no ad-hoc process assumptions	any depth with labels	needs enough, representative sites; generalisation?; hard to interpret?

Motivation Multimodal remote sensing



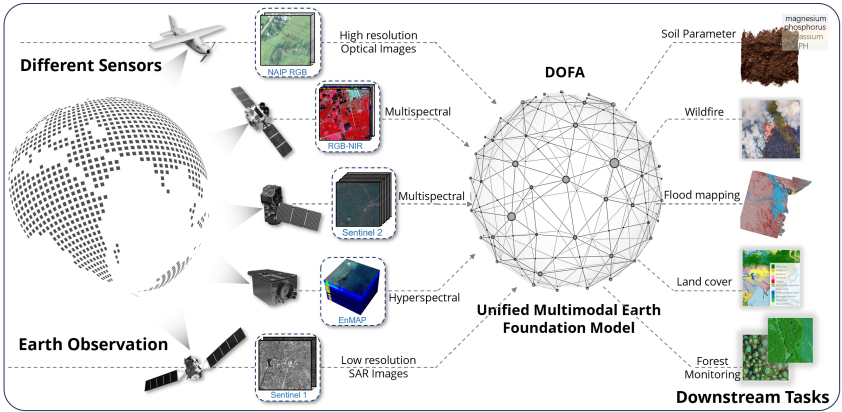
Optical: dry vs wet soil; green vs water-stressed vegetation



SAR scattering types (Nurmemet et al., 2025)

Soil and vegetation water change both optical reflectance and SAR backscatter

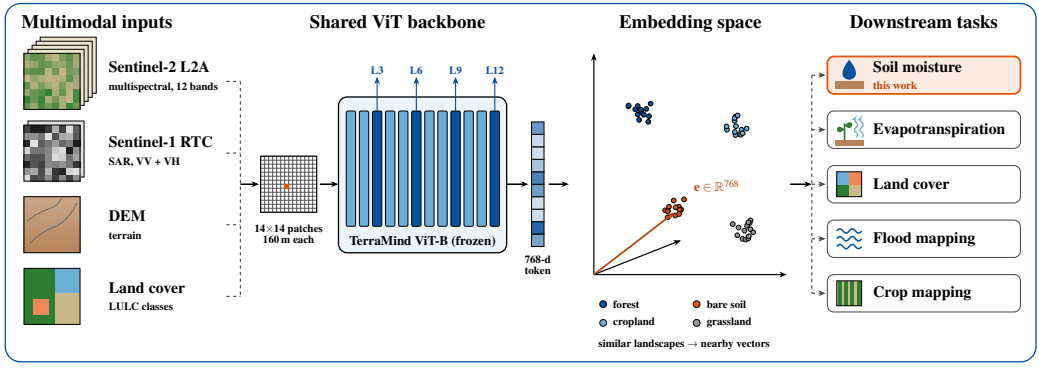
Motivation Geospatial foundation models (GFMs)



DOFA: one model for many sensors and many downstream tasks (Xiong et al., 2024, arXiv:2403.15356)

Once pre-trained on multimodal, unlabelled data, it works for many downstream tasks

Motivation TerraMind embeddings for soil moisture

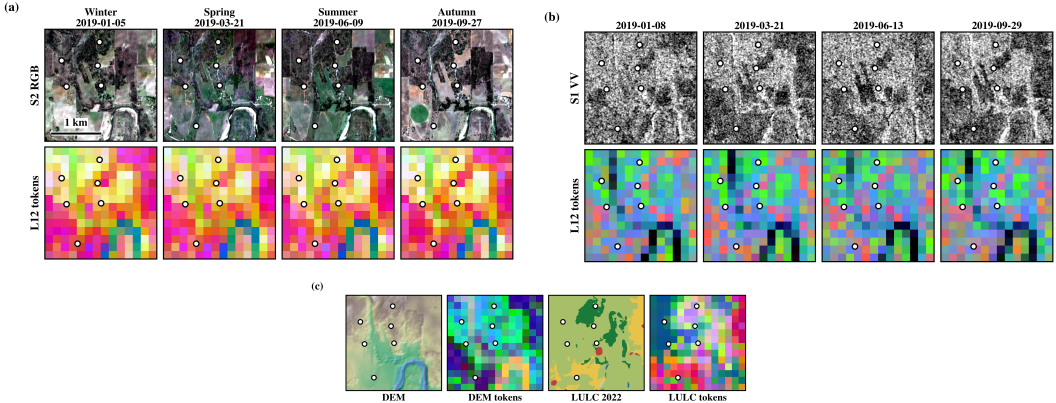


S2, S1, DEM and LULC → frozen TerraMind ViT-B → one 768-d embedding per 160 m patch → downstream tasks

The same embedding can serve many tasks; in this case, soil moisture

Motivation

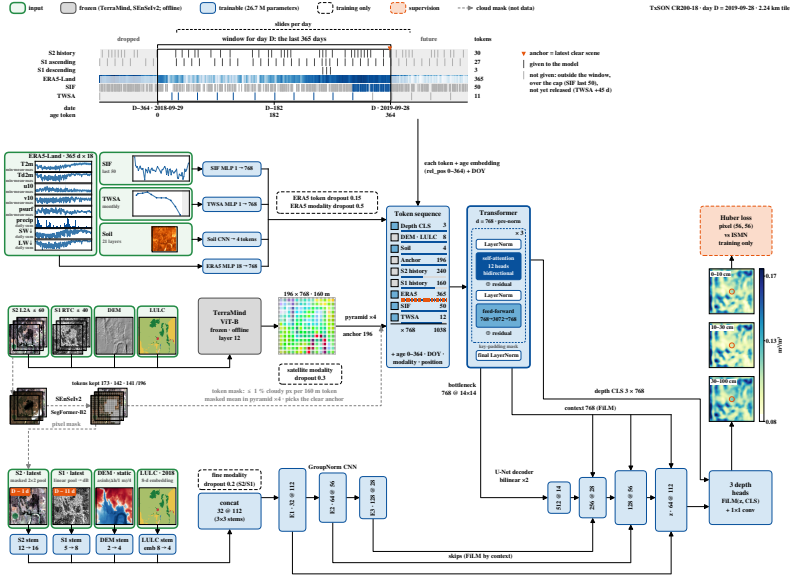
Example: TerraMind embeddings at one site (TxSON, 2019)



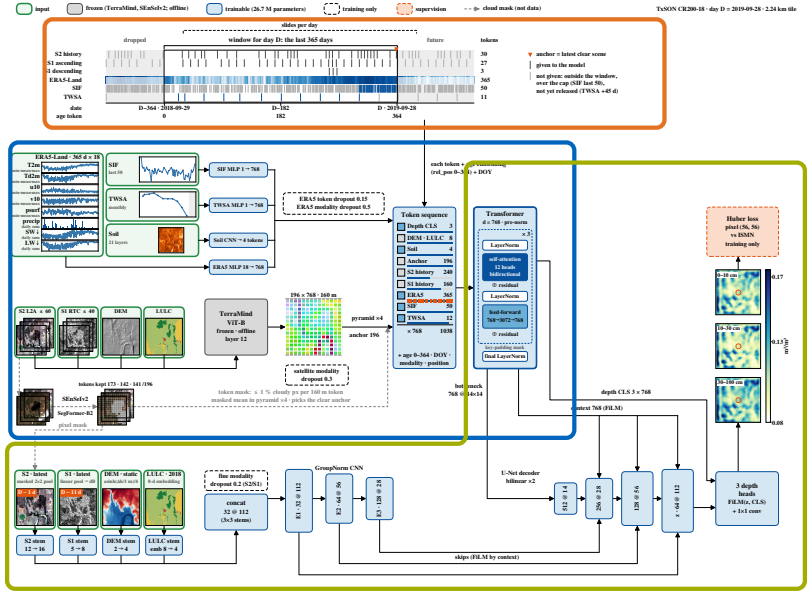
(a) S2 in four seasons, (b) S1 VV, (c) DEM and LULC, each with its TerraMind L12 tokens

Each 160 m patch → one 768-d vector; colours show its first 3 principal components

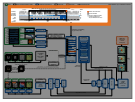
Method Model architecture

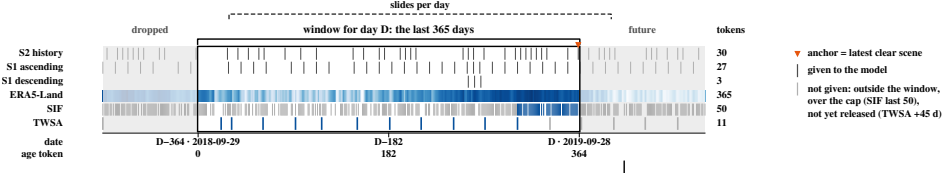


Method Model architecture

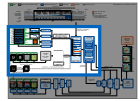


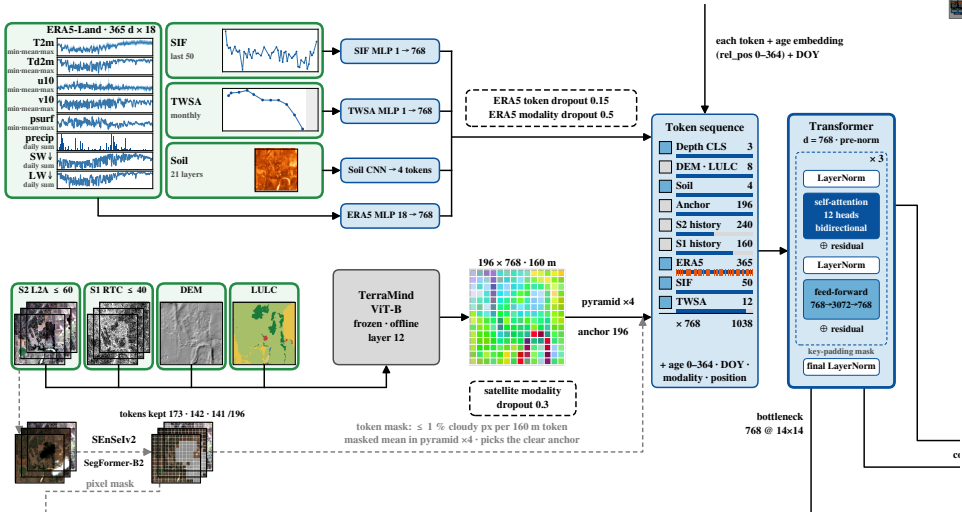
Method Inputs: what the model sees for day D



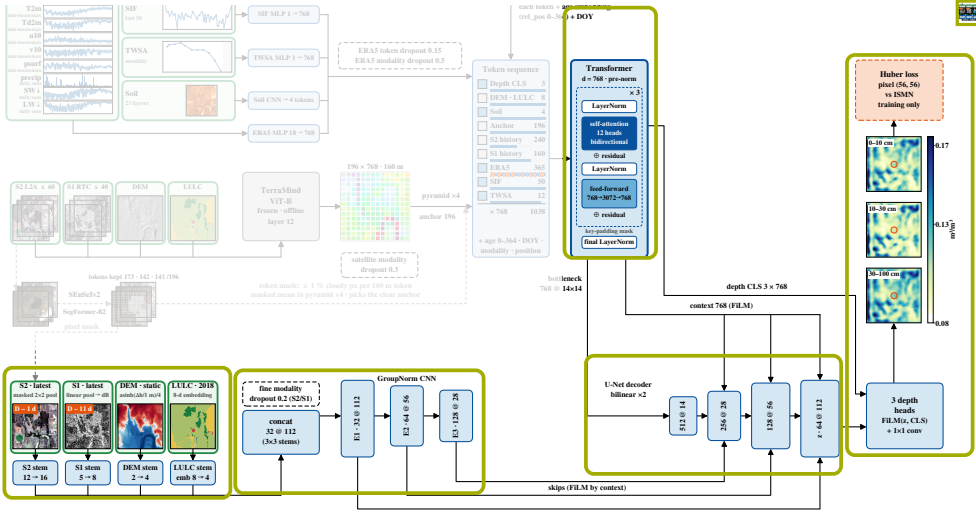


Method Coarse path: tokens → transformer

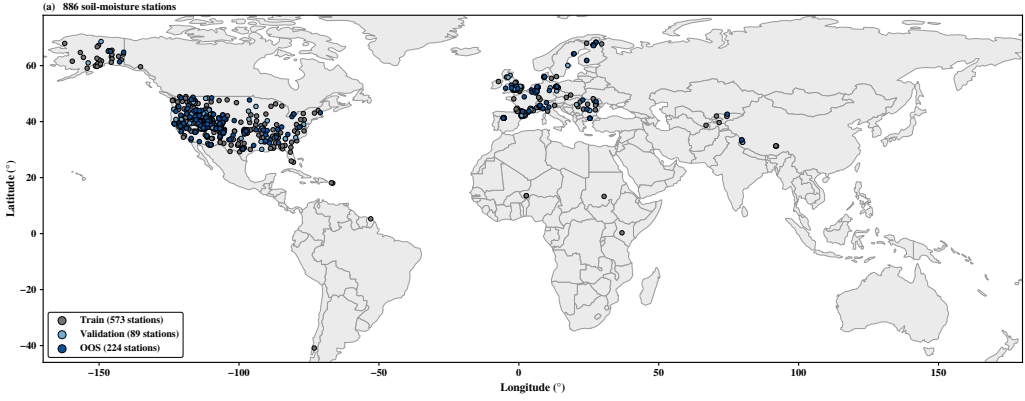




Method **Fine path: 20 m CNN → U-Net decoder → depth heads**



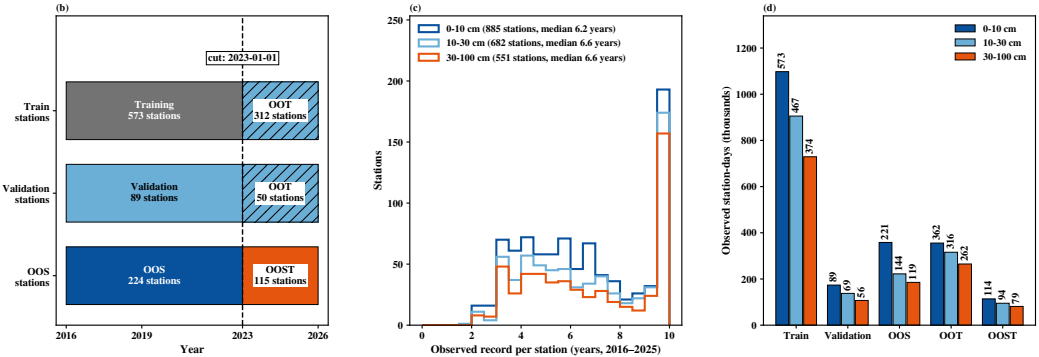
Data In-situ soil moisture: global coverage



886 soil-moisture stations (ISMN and flux sites): train 573, validation 89, out-of-sample (OOS) 224

Dense in North America and Europe; sparse in the tropics and the Southern Hemisphere

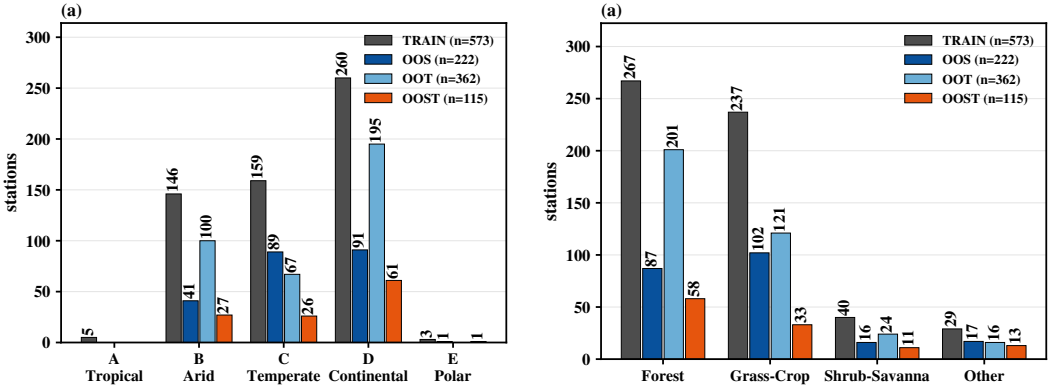
Data Depth coverage, record length and splits



(b) train / validation / OOS stations, time cut 2023-01-01; (c) observed record per station; (d) observed station-days

0–10 cm: 885 stations (median 6.2 yr); 10–30 cm: 682 (6.6 yr); 30–100 cm: 551 (6.6 yr)

Data Climate and land-cover coverage



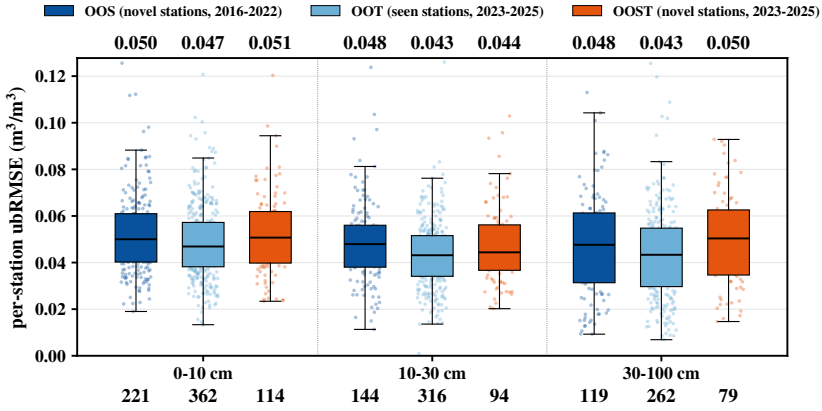
Stations per Köppen climate zone (left) and land-cover class (right), by split

Mostly continental, temperate and arid; forest and grass–crop; almost no tropics or polar

Stations	990 in inventory; **886** with soil moisture (train 573 / val 89 / OOS 224)
Depths	0–10 cm: 885 10–30 cm: 682 30–100 cm: 551 stations
Period	2016–2025; out-of-time test from 2023-01-01
Soil-moisture observations	**5.14 M** observed daily values (2.10 M / 1.68 M / 1.37 M per depth)
Sentinel-2 L2A	272,281 scenes (994 sites); cloud $< 75\,\%$ per scene, cloudy tokens masked
Sentinel-1 RTC	496,835 scenes: 251,310 ascending + 245,525 descending
ERA5-Land	18 variables, 2.46 M station-days (990 stations)
Static / other	DEM, LULC, soil (OpenLandMap), SIF, TWSA

About 5.1 million daily soil-moisture observations at 886 stations, 0–100 cm

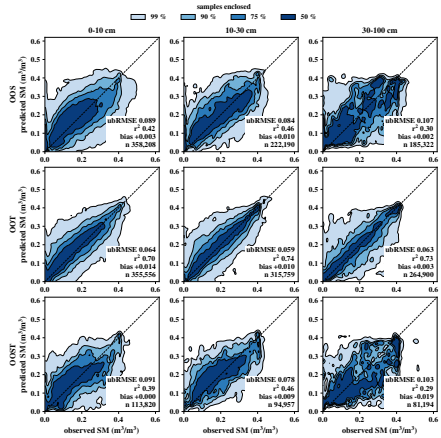
Results Per-station ubRMSE by depth



One point per station; medians above, station counts below

Median per-station ubRMSE 0.043–0.051 m^3/m^3 at every depth and split

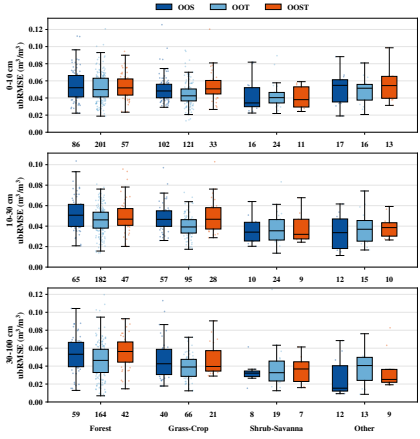
Results Predicted vs observed, by split and depth



All daily samples, smoothed density; shaded bands hold 50/75/90/99 % of samples; rows OOS, OOT, OOST

New years at known sites (OOT, r^2 0.70–0.74) fit far better than new sites (r^2 0.29–0.46)

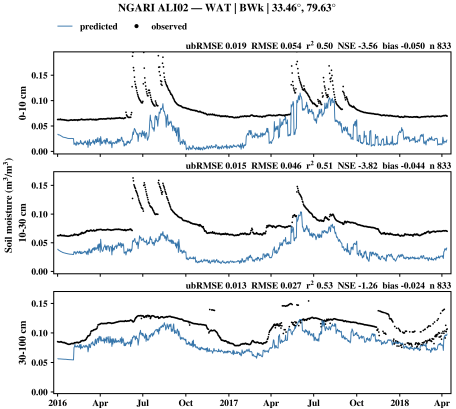
Results Per-station ubRMSE by land cover



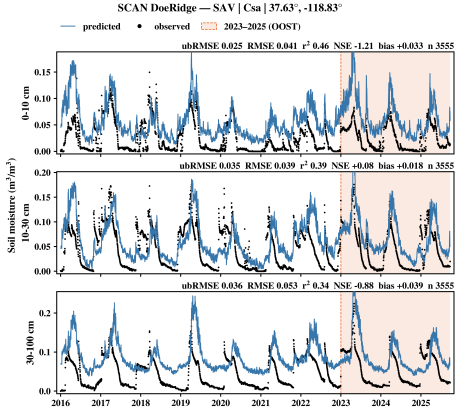
Rows: depths; groups: land-cover class; colours: OOS, OOT, OOST

Similar error across land cover; forest slightly highest, shrub–savanna lowest

Results Sites where all three depths match



NGARI ALI02, Tibetan Plateau (best)

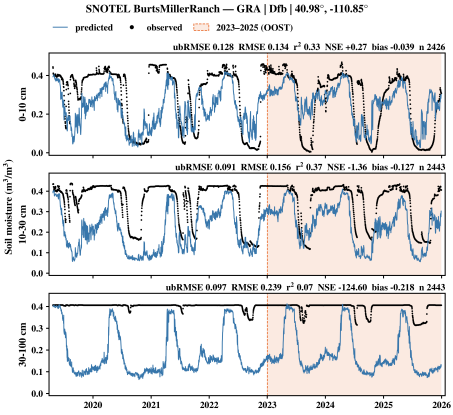


SCAN Doe Ridge, California

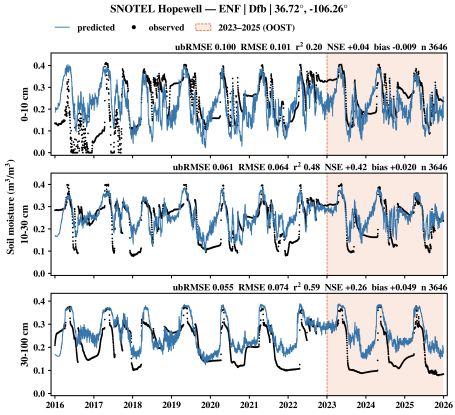
OOS stations observed at all three depths, lowest ubRMSE; blue predicted, black observed

Dynamics right at all depths (ubRMSE 0.013–0.036, r 0.57–0.73); level off by up to 0.05

Results Sites where the depths do not match



SNOTEL Burts Miller Ranch, Utah

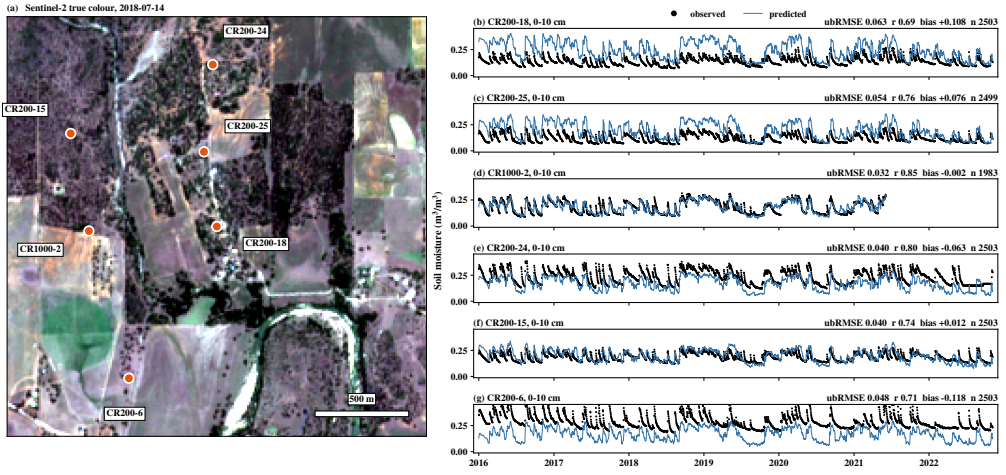


SNOTEL Hopewell, New Mexico

OOS stations observed at all three depths, highest ubRMSE; shaded = 2023–2025 (OOST)

Snowmelt peaks timed right; summer dry-down wrong: too dry deep at Burts Miller, too wet at Hopewell

Results TxSON: six stations inside one 2.2 km tile



(a) Sentinel-2 true colour with the six stations; (b–g) 0–10 cm, predicted vs observed

Timing right at every station (r 0.69–0.85); offsets remain (bias −0.12 to +0.11)